

Dear Editor-in-Chief,

Please consider the enclosed research article, **“Proxy-Based Urban Change Prediction under Spatial Generalization: Improving Tail Reliability with Lightweight Calibration,”** for publication in *Transactions in GIS*.

This study addresses a critical methodology gap in remote-sensing-enabled urban monitoring: the severe mismatch between reported model performance and real-world reliability when predictions are transferred to unseen spatial units. Using the CivitasGrid-TLP dataset (2015–2024), we evaluate proxy-based urban change prediction from WorldPop and VIIRS under a leakage-aware GroupSplit(grid_id) protocol. We benchmark multiple machine learning regressors and introduce a lightweight post-hoc calibration method tailored for spatial regression.

Our main findings deliver key insights for the GIScience community:

- Δ WorldPop is substantially more predictable than Δ VIIRS, whose year-to-year differencing amplifies observational noise and leaves weak signal under spatial generalization.
- Tail errors (rare local shocks) dominate practical model failure, meaning that conventional average metrics (e.g., global R^2) severely understate forecasting risks in vulnerable areas.
- Our proposed RMSE-anchored contrastive affine calibrator consistently improves tail reliability across diverse baselines, providing an effective, model-agnostic tool to safeguard spatial predictions against long-tailed anomalies.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. The author declares no competing interests and received no external funding for this work.

Thank you for your time and consideration.

Sincerely,

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